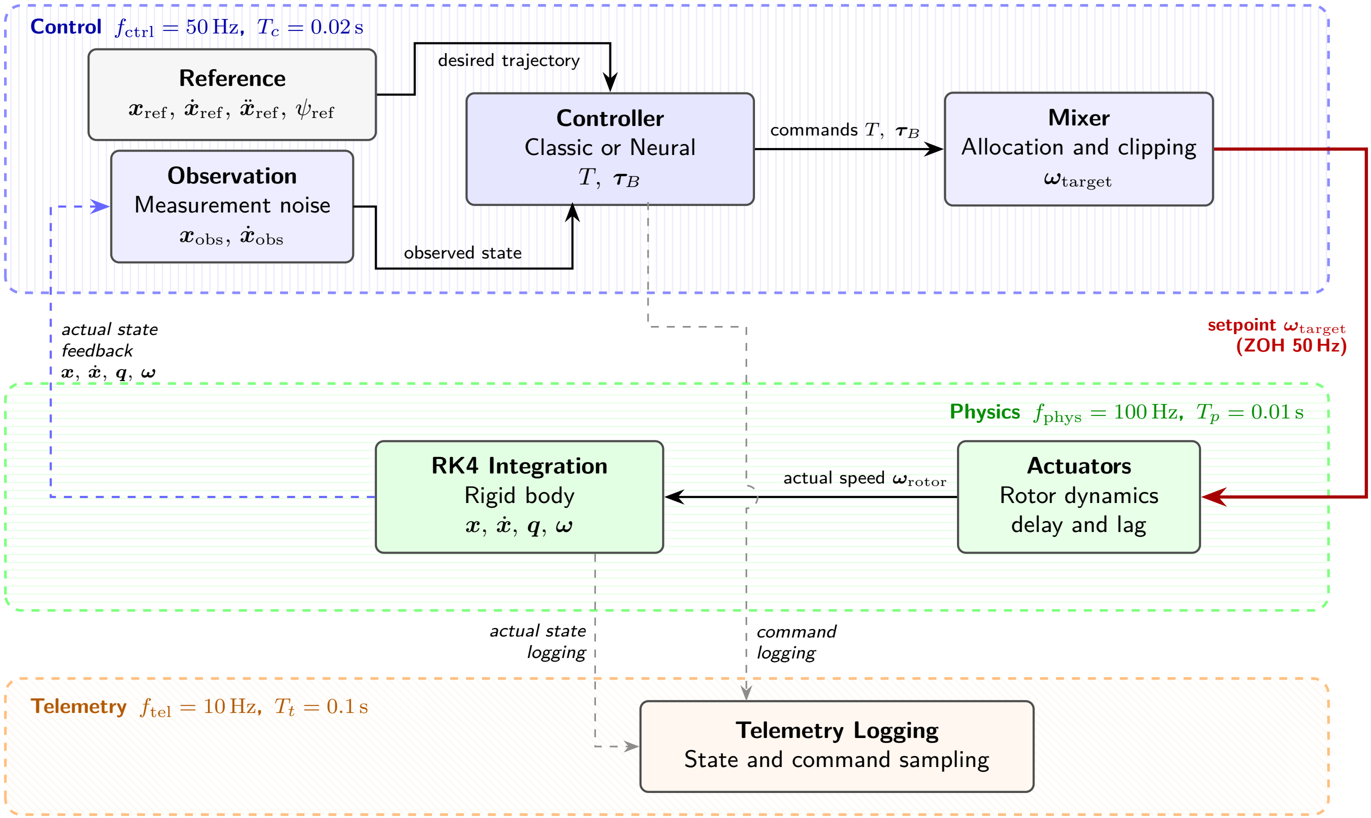
Control $f_{\mathrm{ctrl}} = 50\,\mathrm{Hz}$, $T_c = 0.02\,\mathrm{s}$
Reference
$\boldsymbol{x}_{\mathrm{ref}},\ \dot{\boldsymbol{x}}_{\mathrm{ref}},\ \ddot{\boldsymbol{x}}_{\mathrm{ref}},\ \psi_{\mathrm{ref}}$
desired trajectory
Controller
Classic or Neural
$T,\ \boldsymbol{\tau}_B$
commands $T,\ \boldsymbol{\tau}_B$
Mixer
Allocation and clipping
$\boldsymbol{\omega}_{\mathrm{target}}$
Observation
Measurement noise
$\boldsymbol{x}_{\mathrm{obs}},\ \dot{\boldsymbol{x}}_{\mathrm{obs}}$
observed state
actual state
feedback
$\boldsymbol{x},\ \dot{\boldsymbol{x}},\ \boldsymbol{q},\ \boldsymbol{\omega}$
setpoint $\boldsymbol{\omega}_{\mathrm{target}}$
(ZOH 50 Hz)
Physics $f_{\mathrm{phys}} = 100\,\mathrm{Hz}$, $T_p = 0.01\,\mathrm{s}$
RK4 Integration
Rigid body
$\boldsymbol{x},\ \dot{\boldsymbol{x}},\ \boldsymbol{q},\ \boldsymbol{\omega}$
actual speed $\boldsymbol{\omega}_{\mathrm{rotor}}$
Actuators
Rotor dynamics
delay and lag
actual state
logging
command
logging
Telemetry $f_{\mathrm{tel}} = 10\,\mathrm{Hz}$, $T_t = 0.1\,\mathrm{s}$
Telemetry Logging
State and command sampling